

Chapter – 2: Diversity in the Living World

Trees stand in the Sun and give shade to others. Their fruits are also for others. Likewise, good people bear all hardships and bring welfare to others. They give to others whatever they have earned. – Wise saying

- Pleasant morning after yesterday's rain
- Dr. Raghu, Maniram *chacha* – invited to school by Madam Sulekha – science teacher – facilitate nature walk
- Dr. Raghu – scientist – nearby Research Lab
- Maniram *chacha* – elderly person – nearby comm. – expert – mimicking bird calls – also brilliant – identify – variety of plants, animals
- Prepare – nature walk – Dr. Raghu – informs students – objective – experience beauty, variety – nature
- Students – excited to join – curious to learn
- Teacher – advise students – carry – notebook, pen, water bottle
- All of them – start walking – start exploring
- Dr. Raghu – advise – notice the smells – respect all living creatures – obs. without disturbing
- Maniram *chacha* – advise – not only obs. – BUT also – listen – diff. sounds
- Students – came across –
 - Variety of plants – grasses, bushes, large trees
 - Variety of birds – sitting on trees – butterflies – jumping flower to flower – monkeys – jumping tree to tree
- Students – record obs. – notebooks – discuss with – Maniram *chacha* and Dr. Raghu
- Students can hear – chirping of birds
- Dr. Raghu – informs – each bird – unique chirp – example of diversity
- Dr. Raghu – requests Maniram *chacha* – mimic calls – some birds
- Students – very enthusiastic – start copying

Diversity in Plants and Animals Around Us

- Activity –
 - Plan – nature walk – nearby park, forest
 - **Obs.** – diff plants, animals, insects, birds – also note – weather
 - Collect – diff. leaves, flowers (fallen) – **create** – scrapbook
 - Imp. – do not pluck leaves, flowers
 - **Record** obs. – features – stems, leaves, flowers OR anything interesting
- Plants – variety of features –
 - Tall / short, hard / soft stem
 - Diff. shapes – leaves – diff. – arrangement – stem, branches
 - Flowers – vary – colour, shape, scent
- Create list – animals – record – places they live in, food they eat
- Some animals – live on land – some – live on trees
- Fish – live in water – frogs – live on land and water
- Animals – variety of food – variety of movements
- Sketch – plants, animals – scrapbook – leaves, flowers, feathers, etc. – note down details
- Travelling to school – obs. surroundings
- Activity –
 - Close eyes – 30 sec. – think of 1 plant, 1 animal
 - Draw – blackboard – are they diff. from each other – some – may be common

- BUT – many more varieties – out there
- Variety – plants, animals – particular region – contribute to **biodiversity** – that region
- Each member – diff. role
- Trees – provide food, shelter – animals – help spread seeds – both – dependent on each other

How to Group Plants and Animals?

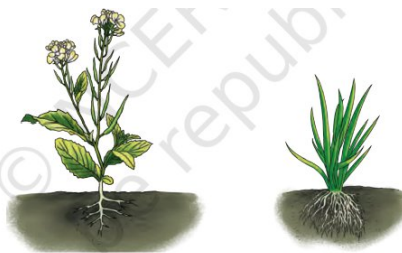
- Arrange your books, notebooks – grps. – arranging – helps – better organize
- World – around us – variety – plants, animals – diff. features
- Grp. them – basis – similarities, differences
- Activity –
 - Collect pictures – various plants, animals – paste – diff. cards
 - Divide class – grps. – 5-6 students
 - Pool cards – prepared in adv.
 - Recall features – plants, animals – **grp.** them – common features
- Basis – diff. grps. – vary
- Reason – diff. students – diff. common features
- Some students – height of plants – others – presence / absence of flowers
- Animal grps. – varied features – what they eat, where they live, their colour
- Grouping – easier to study – plants, animals

How to grp. plants?

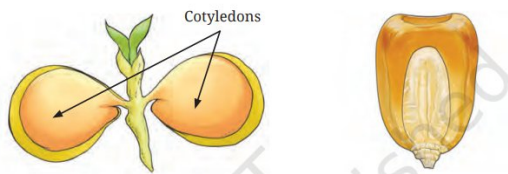
- Plants – show var. – features – stems, leaves, flowers, etc.
- Stems – vary – thickness, height, hardness
- Leaves – vary – shape, colour, size, arrangement
- Earlier classes – learnt – plants – grouped – herbs, shrubs, trees – based on height, types of stem
- Study – features – more details
- Activity –
 - Go on nature walk – more interesting obs.
 - Look closely – height of plants – shorter, same height, taller than you
 - Stem – brown, green – bend gently – stiff OR bent easily – take care – stems – do not break
 - Also obs. – branches of plants – arise from where – close to ground OR higher on stem
- What differences – obs. among – herbs, shrubs, trees
- How to grp. – herbs, shrubs, trees – above obs.
- Some plants – really tall – hard, thick, brown woody stems
- Branches – higher up the stem, away from ground – **trees** – mango tree
- Some plants – not as tall – brown woody stems – start branching – close to ground
- Stems – hard – BUT – not as thick as tree – **shrubs** – rose plant
- Some plants – weak stems – need support – climb, grow – **climbers**
- Some plants – creep on ground – **creepers**
- What other features – grp. them – another activity –
 - Obs. – leaves – diff. plants – variation – shape, structure
- May obs. – thin lines – surface of leaves – **veins**
- Pattern of veins – **venation** – many differences
- Some leaves – net-like pattern – both sides – thick middle vein – **reticulate venation** – hibiscus
- Some leaves – veins – run parallel – **parallel venation** – banana leaves



- Let's **explore** – roots – activity –
 - Visit – open area – wild plants – grow
 - Use *khurpi* (trowel) – dig out – diff. herbs – without damaging roots
 - Wash roots – obs. them – after that – make sure – replant
- Carefully obs. – roots – mustard plant – 1 main root – small side roots
- Main root – **taproot** – another example – taproots – hibiscus
- Roots – common grass plant – bunch – similar-size thin roots – **fibrous roots**



- Any rel. b/w – type of venation and type of roots – activity –
 - Collect saplings – 5 common plants – lemongrass, marigold, *sadabahar* (periwinkle), etc.
 - Before planting – obs. – venation, roots – record obs.
- Sadabahar* plant – taproot – reticulate venation – also chickpea (*chana*)
- Lemongrass – fibrous roots – parallel venation – also wheat
- Rel. – taproot – always pair with – reticulate venation – AND – fibrous roots – always pair with – parallel venation
- Any rel. – seeds, roots, venation – activity –
 - Soak – chickpea, maize seeds – 2-3 days
 - Remove – seed coat – chickpea – obs. – structure – both seeds
- Chickpea seeds – split – 2 parts – each part – **cotyledon**
- Seeds – 2 cotyledons – **dicotyledons (dicots)**
- Maize – single thin cotyledon
- Seeds – 2 cotyledon – **monocotyledons (monocots)**
- Dicot plants – reticulate venation – taproots
- Monocot plants – parallel venation – fibrous roots



How to grp. animals?

- Like plants – animals – significantly diff.
- Previous act. – some basis – grp animals – explore more
- Activity –
 - Recorded – movements of animals – how they move from one place to another
 - Think – types of movement
 - No. of animals – shown in figure
 - Note down – ways – animals move – name body parts – used for movement

- Diff. animals – diff. movements – fly, run, crawl, walk, hop, jump
- Use diff. body parts – moving – use wings, legs, other parts
- Many animals – differ – shape, size, structure, colour, other features
- Some features – grp. animals – various ways
- Like plants – grouping animals – imp. – understand diversity
- **Janaki Ammal** – Indian botanist – dedicated – env. work – helped – document, preserve – India's rich plant biodiversity
- Key role – Save Silent Valley movement
- Head – Botanical Survey of India – initiated programmes – document – plant diversity

Save Silent Valley movement

- Real story – forest – Palakkad dist., Kerala
- Preserve – untouched beauty – moist evergreen forest – rich biodiversity
- Silent valley – now famous – preserved by – common people – not even local resident
- Battle against – hydroelectric dam – Kunthipuzha river – 10 yrs.
- That time – people used – all possible means – letters, articles, appeals in courts, etc. – movement very successful

Plants and Animals in Diff. Surroundings

- Diff. animals – live in – diff. surroundings
- Movement – depend on surroundings
- Fishes – live in water – streamline bodies – fins for movement
- Goats – live in grassy areas – legs for movement
- Sizes, shapes – differ from each another
- Activity –
 - List names – animals, plants – obs. – diff. regions
- Discuss obs. – with classmates
- Plants, animals – found – 1 region – diff. from – another region
- Alex – cactus plants – thick, fleshy stems – deserts, Rajasthan
- Maya – deodar trees – conical shape, flexible, sloping branches – Himalayas, Himachal Pradesh
- Deserts – very little water – hot in day – cold at night – plants and animals – tolerate, survive – both climates – fleshy stem – store water
- Himalayas – freq. snowfall – trees – ability – slide off snow – conical shape
- Hot desert camel –
 - Long legs, wide hooves – helps in walking on sand – keeps away from sand
 - Humps – 1 each – helps store food – scarce in desert
- Cold desert camel –
 - Short legs, height – helps in walking easily on mountains
 - Humps – 2 each – shrink in late winters – food scarcer around this time
 - Long hairs – head to neck – survive cold winters
- Other students – obs. more things
- Kashi – Rajasthan – camels – small amt. – urine, sweat – dung is dry
- Maya – Nilgiris – plants – bright flowers – **rhododendrons** – shorter heights, smaller leaves
- Pema – Sikkim – rhododendrons – taller
- Sagar – Andaman, Nicobar Islands – huge whale, colourful fish – streamline body – helps in swimming
- Plants, animals – particular region – special features – survive there – **adaptations**

- Place – plant, animals live – **habitat**
 - Sea turtle – ocean
 - Camel – hot, cold desert
 - Rhododendron - mountains
- Habitat – provides everything – survival
- Plants, animals – diff. types – share habitat

- **Salim Ali** – travelled India – obs. diversity – birds
- Documented – list of birds – their routes, habitats
- Recorded – regions – high density – took measures – conserve them
- Keoladeo National Park – Bharatpur, Rajasthan
- Ranganathittu Bird Sanctuary – Mandya, Karnataka
- Birdman of India

- Animals, plants – grp. them – based on habitat
- Plants, animals – live on land – **terrestrial** – forest, deserts, grasslands, mountains
- Plants, animals – live in water – **aquatic** – ponds, lakes, rivers, oceans
- Some animals – frogs – live in both – water, land – **amphibians**
- Habitats – change / damaged – adverse effects on env.